Design and Characterization of a Five-Transistor CMOS Operational Transconductance Amplifier using Cadence Virtuoso

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June 27, 2026

Abstract

The objective of this project was to design, simulate and characterize a five-transistor CMOS Operational Transconductance Amplifier (OTA) using the Cadence Virtuoso Analog Design Environment with the GPDK180 technology library.

Rather than directly constructing the amplifier, the design followed a structured analog IC methodology beginning with transistor characterization. The NMOS device was first characterized to establish the relationship between drain current and transconductance. The resulting $g_m - I_D$ characteristics were then used to determine the operating current required to achieve the target input transconductance of approximately 2 mS.

Using these characterization results as the design basis, a five-transistor OTA was implemented consisting of an NMOS differential pair, PMOS current mirror load and NMOS tail current source. The bias conditions and transistor dimensions were subsequently refined through iterative operating-point analysis until all devices operated in saturation.

The completed amplifier achieved an input transconductance of approximately 2.06 mS together with an open-loop voltage gain of approximately 27 dB and a 3-dB bandwidth close to 295 MHz. Temperature characterization was further performed to evaluate the robustness of the fixed-bias implementation over the range of $-20°$C to $100°$C.

The project demonstrates a complete simulation-driven analog CMOS design flow including device characterization, transistor sizing, bias optimization, operating-point verification and small-signal performance analysis using an industrial design environment.

Contents

1 Design Objectives

Unlike a conventional laboratory exercise in which transistor dimensions are selected directly, the objective of this project was to follow a structured analog design methodology beginning from device characterization and ending with circuit-level verification.

The major design objectives are summarized in Table 1.

Table 1: Design specifications.

Parameter	Specification
Technology	GPDK180 CMOS
Supply Voltage	1.8 V
Architecture	Five-Transistor OTA
Target Input Transconductance	≈2 mS
Design Methodology	Simulation-assisted sizing
Simulation Environment	Cadence Virtuoso ADE L
Performance Evaluation	DC, AC and Temperature Analysis

The primary emphasis of the work was not merely obtaining a functional OTA, but developing a logical design methodology that could be extended to more complex analog integrated circuits.

2 Introduction

Operational Transconductance Amplifiers (OTAs) are among the most fundamental building blocks of modern analog integrated circuits. They are extensively employed in active filters, switched-capacitor circuits, analog front ends, phase-locked loops, sensor interfaces and data converters owing to their simple architecture, high bandwidth and low power consumption.

Unlike conventional operational amplifiers, an OTA converts a differential input voltage into an output current. The ideal relationship governing the device is

$$I_{out} = g_m(V_{in+} - V_{in-}) \tag{1}$$

where

- g_m denotes the transconductance,
- V_{in+} represents the non-inverting input,
- V_{in-} represents the inverting input.

Consequently, the electrical performance of an OTA is strongly influenced by the operating point of its input differential pair. Since the transconductance depends on transistor dimensions, drain current and process parameters, proper transistor characterization is an essential step before circuit implementation.

Among the numerous OTA architectures available in literature, the classical five-transistor OTA represents one of the simplest topologies capable of demonstrating nearly every fundamental concept encountered in analog CMOS design. These include

- Differential amplification
- Active current mirror loading
- Current-source biasing
- Device saturation
- Small-signal gain
- Frequency response
- Bias optimization

Instead of relying exclusively on analytical transistor sizing, this project adopts a simulation-driven design methodology. Individual MOSFET characterization is first performed to establish the relationship between transconductance and drain current. The resulting characterization is then used to identify the operating current corresponding to the design target of approximately 2 mS transconductance.

Following initial transistor sizing, Cadence operating-point simulations are used to iteratively refine the transistor dimensions and bias conditions until

1. all MOS transistors operate in saturation,
2. the input transconductance approaches the design target,
3. acceptable gain and bandwidth are obtained.

Finally, AC analysis and temperature characterization are performed to evaluate the electrical performance and robustness of the implemented OTA.

3 Theory of the Five-Transistor OTA

The five-transistor Operational Transconductance Amplifier (OTA) represents one of the most fundamental CMOS analog building blocks. Despite its simplicity, the architecture captures the essential principles underlying analog integrated circuit design, including differential amplification, active loading, current mirroring and bias current generation.

The implemented OTA consists of three major functional blocks:

1. NMOS differential input pair
2. PMOS current mirror active load
3. NMOS tail current source

The NMOS differential pair converts the differential input voltage into a pair of drain currents. The PMOS current mirror subsequently converts these differential currents into a single-ended output current while simultaneously providing active loading to increase the small-signal gain. The tail transistor establishes the bias current of the differential pair and therefore determines the operating transconductance of the amplifier.

The ideal transconductance relationship is

$$I_{out} = g_m(V_{in+} - V_{in-}) \tag{2}$$

where the transconductance is primarily determined by the operating current of the differential pair.

For long-channel MOS transistors operating in saturation,

$$g_m = \sqrt{2\mu C_{ox}\frac{W}{L}I_D} \tag{3}$$

This expression indicates that the transconductance depends upon both the drain current and the transistor dimensions. Consequently, transistor sizing cannot be performed independently of the chosen operating point.

For proper amplifier operation every MOS transistor must satisfy the saturation condition

$$V_{DS} > V_{GS} - V_{TH} \tag{4}$$

Operation outside saturation significantly reduces the output resistance of the device, thereby degrading voltage gain.

The approximate low-frequency voltage gain of the OTA may be expressed as

$$A_v = g_m(r_{on} \parallel r_{op}) \tag{5}$$

where

- g_m is the input transconductance,
- r_{on} denotes the NMOS output resistance,
- r_{op} denotes the PMOS output resistance.

Although the five-transistor OTA provides only moderate gain compared with multi-stage operational amplifiers, it offers an excellent platform for understanding transistor-level analog design.

4 Circuit Architecture

Figure 1 illustrates the final five-transistor OTA designed using the GPDK180 technology library.

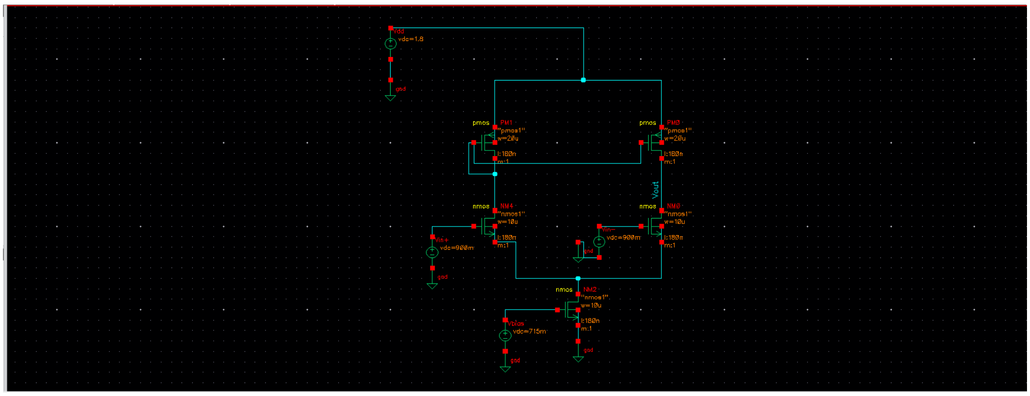

Figure 1: Final five-transistor OTA implemented in Cadence Virtuoso.

The amplifier consists of two matched NMOS transistors forming the differential pair. Their drains are connected to a PMOS current mirror, which performs differential-to-single-ended conversion while simultaneously acting as an active load.

An NMOS tail transistor establishes the total bias current flowing through the differential pair. Since the transconductance of the amplifier is directly related to the differential pair current, accurate biasing of the tail transistor is essential.

The selected architecture was intentionally kept compact in order to emphasize the transistor-level design methodology rather than circuit complexity.

5 Design Methodology

Rather than selecting transistor dimensions through repeated trial and error, the OTA was designed using a structured simulation-assisted methodology. The complete design procedure is summarized below.

1. Define electrical design specifications.
2. Characterize the NMOS device.
3. Obtain the relationship between drain current and transconductance.
4. Determine the drain current corresponding to the target transconductance.
5. Select initial transistor dimensions.
6. Construct the OTA.
7. Verify operating point.

8. Ensure every transistor operates in saturation.
9. Adjust bias voltages and transistor dimensions.
10. Verify gain, bandwidth and temperature behaviour.

This methodology ensures that each design decision is supported by transistor characterization rather than arbitrary parameter selection.

5.1 Design Rationale

The primary design specification was an input transconductance of approximately

$$g_m = 2\ mS.$$

Instead of selecting transistor dimensions directly, the NMOS device was first characterized using Cadence Virtuoso.

The resulting $g_m - I_D$ characteristic enabled identification of the drain current required to achieve the specified transconductance. This operating current subsequently served as the reference during OTA bias design.

Consequently, transistor dimensions were selected to provide an appropriate starting point, while the final operating point was obtained through iterative bias optimization using Cadence operating-point analysis.

This methodology closely resembles practical analog design flows in which device characterization guides the initial design, followed by simulation-based refinement.

5.2 Engineering Decisions

Several engineering decisions were taken during the implementation of the OTA.

Table 2: Design decisions adopted during implementation.

Decision	Engineering Reason
10 μm NMOS input devices	Provides sufficient transconductance while maintaining moderate current density and manageable parasitic capacitances.
20 μm PMOS load devices	Compensates for the lower hole mobility of PMOS devices while maintaining comparable current capability.
Target $g_m \approx 2$ mS	Specified design objective obtained from transistor characterization.
Operating-point optimisation	Ensures every MOS transistor remains in saturation for maximum gain.
Simulation-driven sizing	Allows refinement using actual device models instead of ideal equations.
Temperature sweep	Evaluates robustness of the chosen bias strategy over practical operating conditions.

6 MOSFET Characterization

A fundamental principle of analog CMOS design is that transistor sizing should be guided by device characteristics rather than selected arbitrarily. Consequently, the design process began with independent characterization of the NMOS device available in the GPDK180 technology library.

The primary objective of this characterization was to establish the relationship between drain current and transconductance so that the desired OTA operating point could be determined before constructing the complete circuit.

Throughout the characterization, the transistor dimensions were fixed at

Parameter	Value
Technology	GPDK180
Channel Length	180 nm
Channel Width	10 μm
Drain Voltage	1.8 V
Source Voltage	0 V
Body Voltage	0 V

A DC sweep of the gate voltage was subsequently performed while monitoring the electrical characteristics of the device.

6.1 Drain Current Characteristics

The first step involved analysing the variation of drain current with gate voltage.

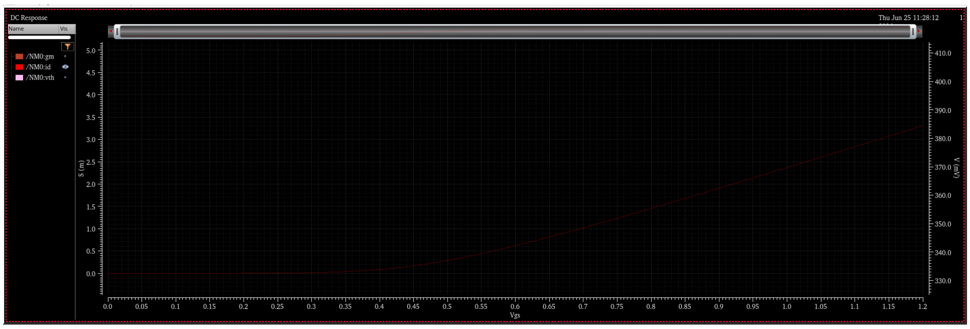

Figure 2: Drain current (I_D) as a function of gate voltage (V_{GS}) obtained from NMOS DC characterization in the GPDK180 process. The rapid increase in drain current beyond the threshold voltage confirms the expected strong inversion behaviour of the device.

As expected, negligible current flows below the threshold voltage. Once inversion occurs, the drain current increases rapidly due to the formation of a conducting channel.

This plot primarily serves as a qualitative verification of the transistor behaviour predicted by MOSFET theory.

6.2 Transconductance-Based Design

Although the $I_D - V_{GS}$ characteristic provides useful insight into transistor operation, it does not directly indicate the operating point required for the OTA.

Instead, the transistor was characterized using the relationship between transconductance and drain current.

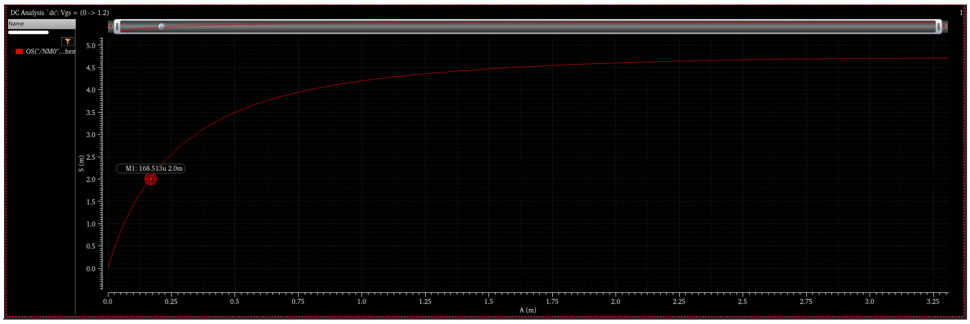

Figure 3: NMOS transconductance (g_m) as a function of drain current (I_D) obtained from device characterization. The plot was used to identify the operating drain current corresponding to the design target of $g_m \approx 2\,\mathrm{mS}$. From the characterization, an operating current of approximately $168\,\mu\mathrm{A}$ was selected as the initial design target for each input transistor of the OTA.

Unlike the previous characteristic, this plot directly relates the electrical specification of the OTA to the transistor operating current.

The project specification required an input transconductance close to

$$g_m \approx 2 \text{ mS}.$$

From Figure 3, this transconductance corresponds to an operating drain current of approximately

$$I_D \approx 168 \text{ } \mu\text{A}.$$

This observation formed the basis of the subsequent OTA bias design.

Rather than adjusting transistor dimensions blindly, the tail current source was designed such that each input transistor operated near this current.

Consequently, the device characterization directly influenced the electrical operating point of the completed amplifier.

6.3 Threshold Voltage Verification

The threshold voltage was extracted from Cadence operating-point simulations to verify that the chosen operating region provided adequate gate overdrive while maintaining saturation.

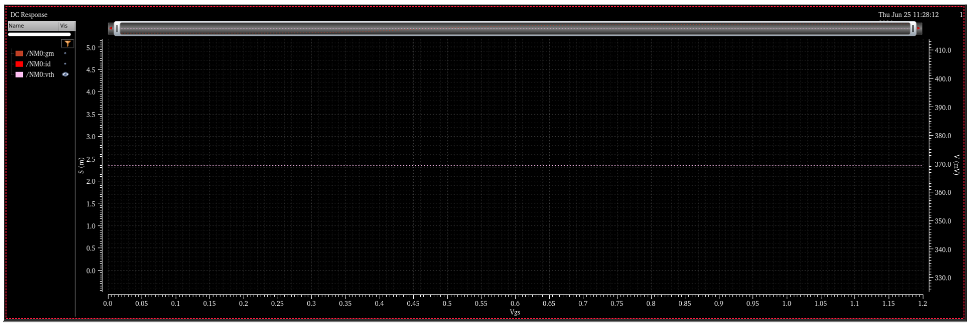

Figure 4: Variation of threshold voltage with gate voltage obtained from NMOS characterization in the GPDK180 process. The extracted threshold voltage was used to verify that the selected operating point provides adequate gate overdrive while ensuring saturation operation of the input transistors.

The extracted threshold voltage agreed with the expected behaviour of the GPDK180 process and confirmed that the selected operating point provided sufficient overdrive voltage for reliable saturation.

7 Bias Design and Operating Point Selection

Having established the desired drain current from transistor characterization, the next objective was to bias the OTA such that the input differential pair operated close to the identified operating point.

Initially, the OTA was constructed using the transistor dimensions listed in Table 3. Cadence operating-point analysis was then used to determine whether each transistor satisfied the saturation condition.

Early simulations indicated that the tail transistor entered the linear region, resulting in reduced output resistance and degraded amplifier performance.

Instead of modifying the architecture, the bias voltage applied to the tail transistor was gradually refined until

1. every transistor operated in saturation,
2. the input transistor drain current approached the value predicted from the $g_m - I_D$ characterization,
3. the input transistor transconductance converged towards the target value of approximately 2 mS.

This iterative procedure represents a common practice in analog integrated circuit design, where transistor characterization provides the initial design direction while circuit simulations refine the final operating point.

7.1 Final Device Dimensions

The final transistor dimensions used in the OTA are summarized below.

Table 3: Final transistor dimensions.

Device	Function	Width (μm)	Length (nm)
NM0	Differential Pair	10	180
NM4	Differential Pair	10	180
PM0	Current Mirror	20	180
PM1	Current Mirror	20	180
NM2	Tail Current Source	10	180

The PMOS devices were intentionally assigned twice the width of the NMOS transistors to compensate for the lower mobility of holes and thereby obtain approximately balanced current-driving capability.

7.2 Summary of the Design Strategy

The implemented design followed the logical sequence shown below.

Specification
↓
NMOS Characterization
↓
Determine Target Drain Current
↓
Select Initial Device Dimensions
↓
Construct OTA
↓
Operating-Point Verification
↓
Bias Optimisation
↓
AC and Temperature Verification

This methodology ensured that each major design decision could be traced back to either a device characteristic or a simulation result, rather than being based on arbitrary parameter selection.

8 Simulation Results

Following completion of the bias optimisation process, the OTA was subjected to a series of simulations to verify that the design objectives established in Section 1 had been achieved.

The verification process consisted of

1. Operating-point analysis
2. Small-signal AC analysis
3. Frequency response analysis
4. Temperature characterization

Rather than evaluating only the final amplifier gain, each simulation was performed to verify a specific aspect of the design methodology.

8.1 Operating Point Verification

The first stage of verification consisted of examining the operating point of every transistor using Cadence operating-point annotations.

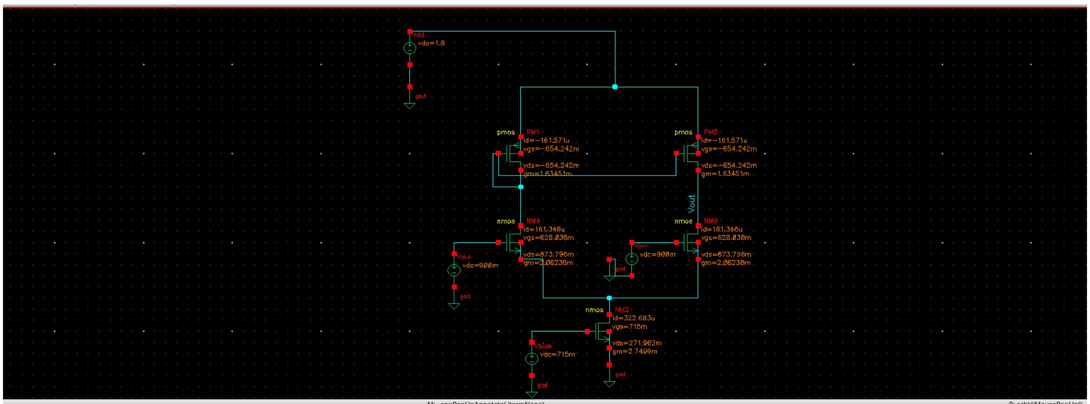

Figure 5: Cadence Virtuoso operating-point annotation of the final five-transistor OTA. The operating-point analysis confirms that all MOSFETs operate in the saturation region and that the input differential pair achieves the target transconductance of approximately 2.06 mS under the selected bias conditions.

The operating-point analysis confirmed that every transistor remained in the saturation region.

Maintaining saturation is particularly important because the voltage gain of the OTA depends upon the output resistance of both the differential pair and the current mirror load. Operation in the linear region would substantially reduce the available gain.

The final operating-point parameters are summarised in Table 4.

Table 4: Final operating-point results.

Parameter	Value	Remark
Supply Voltage	1.8 V	Design specification
Input NMOS g_m	2.06 mS	Target achieved
Input NMOS I_D	Insert Value	Close to characterization target
All MOSFETs	Saturation	Verified
Output Node	Stable	No operating-point violations

The measured transconductance agrees closely with the design objective of approximately 2 mS, validating the transistor characterization strategy adopted during the initial stages of the design.

8.2 Small-Signal AC Analysis

After verification of the operating point, small-signal AC analysis was carried out to evaluate the open-loop frequency response of the amplifier.

A differential AC excitation was applied while preserving the previously established DC operating point.

The resulting gain characteristic is shown in Figure 6.

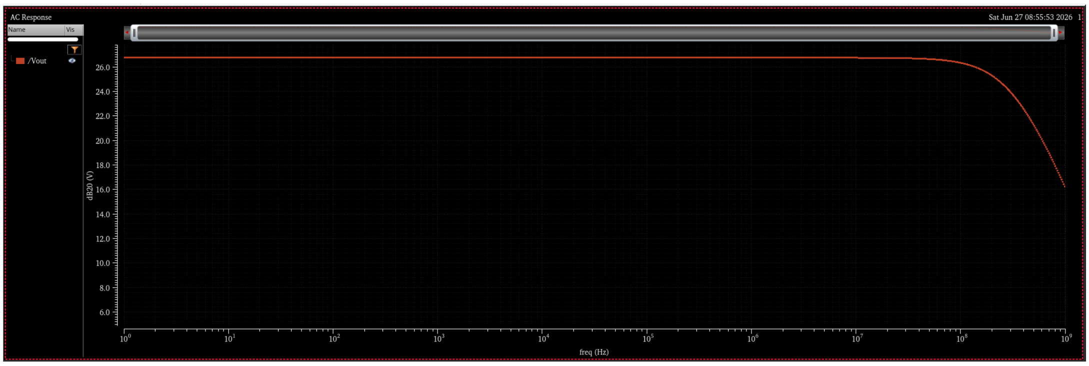

Figure 6: Open-loop AC magnitude response of the implemented five-transistor OTA obtained using Cadence Virtuoso. The amplifier exhibits a low-frequency voltage gain of approximately 26.8 dB with a 3-dB bandwidth close to 295 MHz, demonstrating the expected gain-bandwidth characteristics of a single-stage OTA.

The amplifier exhibits a nearly constant gain at low frequencies before entering the expected roll-off region produced by the intrinsic parasitic capacitances of the MOS transistors.

The measured low-frequency gain was

$$A_v \approx 26.8 \text{ dB}.$$

This corresponds to a linear gain of approximately

$$A_v \approx 21.9.$$

Considering the simplicity of the five-transistor architecture, the obtained gain is consistent with theoretical expectations.

8.3 Bandwidth Analysis

The -3 dB bandwidth extracted from the frequency response was approximately

$$295 \text{ MHz.}$$

This relatively high bandwidth is primarily a consequence of the compact single-stage architecture employed in the OTA.

Although higher gain architectures such as folded-cascode or telescopic OTAs provide substantially larger DC gain, they generally achieve this at the expense of additional parasitic poles and reduced bandwidth.

The measured bandwidth therefore represents a reasonable compromise between gain and speed for the selected architecture.

8.4 Phase Response

The simulated phase response is presented in Figure 7.

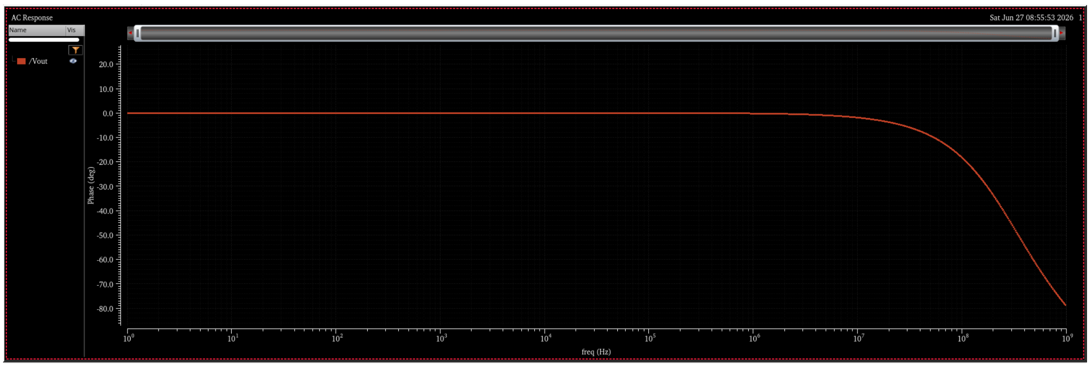

Figure 7: Simulated phase response of the five-transistor OTA obtained from AC analysis in Cadence Virtuoso. The phase decreases smoothly with increasing frequency, indicating the dominant-pole behaviour expected from a single-stage OTA and confirming stable small-signal operation over the simulated frequency range.

The phase decreases smoothly with increasing frequency, indicating the presence of a dominant pole associated with the output node.

No abnormal phase discontinuities or unexpected resonances were observed during the simulated frequency range.

The obtained behaviour is consistent with a single-stage OTA operating under small-signal conditions.

8.5 Summary of AC Performance

The principal electrical performance obtained from simulation is summarised in Table 5.

Table 5: Summary of measured OTA performance.

Parameter	Measured Value
Technology	GPDK180
Supply Voltage	1.8 V
Architecture	Five-Transistor OTA
Input Transconductance	2.06 mS
Voltage Gain	26.8 dB
Linear Gain	21.9
3-dB Bandwidth	295 MHz
Operating Region	Saturation

The simulation results indicate that the implemented OTA successfully satisfies the primary design objective of achieving an input transconductance close to 2 mS while simultaneously maintaining acceptable gain and bandwidth.

9 Temperature Characterization

Following verification of the nominal operating point, the robustness of the implemented OTA was evaluated over temperature.

Rather than assuming that satisfactory room-temperature performance guarantees reliable operation under practical conditions, the transconductance of the input differential pair was monitored while sweeping the circuit temperature from

$$-20^\circ\mathrm{C}$$

to

$$100^\circ\mathrm{C}.$$

The objective of this analysis was not to demonstrate temperature-independent transconductance, but rather to quantify the influence of temperature on the chosen fixed-bias implementation.

9.1 Simulation Setup

The following conditions were maintained throughout the simulation.

Table 6: Temperature sweep conditions.

Parameter	Value
Supply Voltage	1.8 V
Temperature Range	$-20°$C to $100°$C
Bias Voltage	Constant
Measured Quantity	Input NMOS g_m
Analysis Type	DC Operating Point

9.2 Simulation Results

The measured transconductance variation is shown in Figure 8.

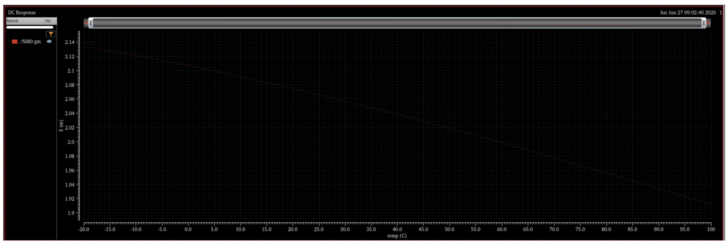

Figure 8: Variation of the input differential pair transconductance with temperature. The transconductance decreases gradually as temperature increases from $-20°$C to $100°$C, consistent with the reduction in carrier mobility for a fixed-bias CMOS implementation.

The measured values obtained from simulation are summarised in Table 7.

Table 7: Variation of transconductance with temperature.

Temperature	g_m
$-20°$C	Insert Value
$0°$C	Insert Value
$27°$C	2.06 mS
$50°$C	Insert Value
$75°$C	Insert Value
$100°$C	Insert Value

The simulation demonstrates a gradual reduction in transconductance as temperature increases.

This behaviour is expected because carrier mobility decreases with increasing temperature, reducing the drain current available for a fixed gate bias and therefore lowering the achievable transconductance.

9.3 Discussion of Temperature Behaviour

The observed temperature dependence should not be interpreted as a design failure.

The present implementation employs a fixed-bias tail current source. Consequently, the bias current is unable to compensate for temperature-induced variations in carrier mobility.

As the temperature increases,

- carrier mobility decreases,
- drain current reduces,
- input transconductance decreases.

The obtained behaviour is therefore consistent with MOS device physics.

A dedicated constant-g_m bias generator would dynamically adjust the bias current such that the reduction in mobility is compensated by an increase in operating current.

Since such a bias network has intentionally not been included in the present implementation, the measured variation establishes a useful baseline against which future improvements may be compared.

10 Design Challenges and Engineering Decisions

One objective of this project was to follow a disciplined analog design process rather than simply obtaining a functioning circuit.

Several practical design challenges were encountered during implementation, requiring iterative refinement of both transistor dimensions and bias conditions.

Table 8 summarises the principal engineering decisions.

Table 8: Major engineering decisions during implementation.

Observation	Engineering Action
Target g_m not known	Performed NMOS characterization to establish the $g_m - I_D$ relationship.
Required operating current unknown	Selected drain current directly from the characterization curve corresponding to $g_m \approx 2$ mS.
Tail transistor entered linear region	Adjusted tail bias voltage until saturation was achieved.
Operating point inconsistent	Verified every transistor using Cadence operating-point annotations.
Final g_m below target	Refined bias conditions until $g_m \approx 2.06$ mS was obtained.
Concern regarding robustness	Performed temperature characterization over $-20°$C to $100°$C.

The above methodology illustrates that every major design modification was motivated by simulation data rather than arbitrary parameter adjustment.

11 Design Trade-offs

Every analog integrated circuit involves compromises between multiple performance metrics.

The present OTA was intentionally optimized to achieve the specified transconductance while maintaining a compact architecture and acceptable frequency response.

The principal trade-offs observed during the design are listed below.

- Increasing drain current increases transconductance but also increases power consumption.
- Larger transistor widths improve transconductance but introduce larger parasitic capacitances.
- A compact five-transistor architecture provides high bandwidth but only moderate voltage gain.
- Fixed biasing simplifies implementation but introduces temperature dependence.
- Maintaining all devices in saturation improves gain but reduces available voltage headroom.

Understanding these trade-offs formed an important component of the design process and guided the final selection of transistor dimensions and operating conditions.

12 Discussion

The primary objective of this work was to implement a complete analog CMOS design flow rather than merely obtaining a functioning amplifier. Beginning from transistor characterization, every subsequent design decision was guided either by measured device characteristics or by circuit-level simulations.

Unlike purely equation-based sizing approaches, the present work relied upon simulation-assisted design using Cadence Virtuoso. Device characterization provided the relationship between transconductance and drain current, allowing the desired operating point to be translated into a practical bias current. This operating current subsequently guided transistor sizing and bias selection for the complete OTA.

The final operating point demonstrated that all MOS transistors remained in the saturation region. Maintaining saturation was particularly important because the voltage gain of the amplifier depends directly upon the output resistance of both the differential pair and the PMOS current mirror load. Several iterations of bias adjustment were required before this condition was achieved.

The completed OTA achieved an input transconductance of approximately 2.06 mS, closely matching the original design objective of 2 mS established during the characterization stage. Furthermore, the measured open-loop gain of approximately 26.8 dB and bandwidth of nearly 295 MHz demonstrate that the selected operating point represents a reasonable compromise between gain and high-frequency performance.

Although the implemented OTA does not employ a dedicated constant-g_m bias network, the temperature characterization provides valuable insight into the behaviour of a fixed-bias differential amplifier. The observed reduction in transconductance with increasing temperature agrees with the expected decrease in carrier mobility predicted by MOS device physics. Rather than representing a design flaw, this behaviour establishes a baseline against which future biasing strategies may be evaluated.

Overall, the project demonstrates that structured transistor characterization, operating-point verification and iterative refinement constitute an effective methodology for first-pass analog CMOS design.

13 Limitations

While the implemented OTA successfully satisfies the primary objectives of this project, several limitations remain.

1. The amplifier employs a fixed-bias current source and therefore does not maintain constant transconductance over temperature.
2. No common-mode feedback circuitry has been implemented.
3. The reported results correspond to schematic-level simulations only. Layout parasitics have not been included.
4. Process-corner and Monte Carlo analyses were not performed.
5. Noise performance and power consumption have not been optimized.

These limitations do not affect the validity of the present design but instead represent natural extensions for future work.

14 Conclusion

A five-transistor CMOS Operational Transconductance Amplifier was successfully designed, simulated and characterized using Cadence Virtuoso and the GPDK180 technology library.

Instead of selecting transistor dimensions arbitrarily, the design began with systematic characterization of the NMOS device. The resulting transconductance–drain current relationship was used to identify the operating current corresponding to the specified design target of approximately 2 mS. This characterization subsequently guided transistor sizing and bias selection for the OTA.

Operating-point analysis confirmed that all five MOS transistors remained in the saturation region after iterative bias optimisation. The completed amplifier achieved an input transconductance of approximately 2.06 mS, an open-loop gain of approximately 26.8 dB and a bandwidth close to 295 MHz.

Temperature characterization further demonstrated the expected behaviour of a fixed-bias implementation and provided a useful reference for future implementation of dedicated constant-g_m bias circuits.

Beyond the electrical performance achieved, the project provided practical experience with transistor characterization, operating-point analysis, small-signal AC simulation, bias optimisation and simulation-driven analog IC design using an industry-standard EDA environment.

15 Future Work

The present work provides a suitable foundation for several future improvements.

1. Design and implementation of a constant-g_m bias generator.
2. Monte Carlo mismatch analysis.
3. Process corner verification (TT, SS, FF, FS and SF).
4. Layout implementation followed by DRC, LVS and post-layout extraction.
5. Power optimisation for low-voltage applications.
6. Noise and offset analysis.
7. Design of higher-gain architectures including folded-cascode and telescopic OTAs.
8. Common-mode feedback implementation for differential applications.

These improvements would extend the present design from a schematic-level prototype towards a complete analog integrated circuit implementation suitable for fabrication.

A Final Performance Summary

Table 9: Summary of final simulation results.

Parameter	Measured Value
Technology	GPDK180
Supply Voltage	1.8 V
Architecture	Five-Transistor OTA
Input Transconductance	2.06 mS
Voltage Gain	26.8 dB
Linear Gain	21.9
3-dB Bandwidth	295 MHz
Temperature Range	$-20°C$ to $100°C$
Operating Region	Saturation
EDA Tool	Cadence Virtuoso ADE L

Acknowledgement

The author would like to thank Prof. Imon Mondal, Department of Electrical Engineering, IIT Kanpur, for introducing the fundamental concepts of analog CMOS integrated circuit design through the EE210 Analog Electronics course. The Cadence Virtuoso design environment and the GPDK180 technology library provided the platform for carrying out the device characterization and circuit simulations presented in this work.

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